T6A2: Helmholtz equation through its weak form

Equation: $\phi''(x) + k^2\phi(x) = -s(x)$ on $[0,1]$ with $\phi(0) = \phi(1) = 0$, $k = \pi$ and $s(x) = \delta(x - 0.5)$.

1) Weak form

Same as section 9.3 in the book (Eqs. 9.35–9.38). I multiply the equation by $\psi(x)$ with $\psi(0) = \psi(1) = 0$ and integrate from 0 to 1. Then I do integration by parts on $\phi''\psi$. The boundary term is 0 because $\psi(0) = \psi(1) = 0$:

$$\int_0^1 \phi''\psi\,dx = \Big[\phi'\psi\Big]_0^1 - \int_0^1 \phi'\psi'\,dx = -\int_0^1 \phi'\psi'\,dx.$$

The δ gives $\int_0^1 \delta(x - 0.5)\,\psi\,dx = \psi(0.5)$. So the weak form is:

$$\int_0^1 \big(\phi'\psi' - k^2\phi\psi\big)\,dx = \psi(0.5) \qquad \text{for every } \psi \text{ with } \psi(0) = \psi(1) = 0.$$

Only ϕ' is in the weak form, so ϕ has to be continuous but ϕ' can jump.

2) Solving the weak form

I use the basis from section 9.2 (Eqs. 9.12 and 9.13): $u_l = \sqrt{2}\sin(l\pi x)$, $l = 1, 2, \ldots$ They are orthonormal, so $\phi = \sum_l a_l u_l$ with $a_l = \int_0^1 \phi\,u_l\,dx$. I put $\psi = u_l$ in the weak form. For the first term I do integration by parts again, but now the derivative goes to u_l. The boundary term is 0 because $\phi(0) = \phi(1) = 0$, and $u_l'' = -l^2\pi^2 u_l$:

$$\int_0^1 \phi' u_l'\,dx = \Big[\phi\,u_l'\Big]_0^1 - \int_0^1 \phi\,u_l''\,dx = l^2\pi^2 a_l.$$

With $u_l(0.5) = \sqrt{2}\sin(l\pi/2)$, the weak form becomes:

$$\big(l^2\pi^2 - k^2\big)\,a_l = \sqrt{2}\sin\frac{l\pi}{2}.$$

This is true for every solution of the weak form, because I only used that ϕ is continuous and $\phi(0) = \phi(1) = 0$.

For $k = \pi$ and $l = 1$: the left side is $(\pi^2 - \pi^2)a_1 = 0$ and the right side is $\sqrt{2}\sin(\pi/2) = \sqrt{2}$. So $0 = \sqrt{2}$, which is not possible. Therefore, the weak form has no solution.

Reason: $\sin(\pi x)$ already solves $\phi'' + \pi^2\phi = 0$ with $\phi(0) = \phi(1) = 0$, so $k^2 = \pi^2$ is an eigenvalue (Eq. 9.43). The δ at $x = 0.5$ hits this mode at its maximum, so a_1 would have to be infinite (resonance).

3) Solving the equation

For $x \neq 0.5$: $\phi'' + \pi^2\phi = 0$, so on each side ϕ is a mix of $\sin(\pi x)$ and $\cos(\pi x)$.
At $x = 0.5$: I integrate the equation from $0.5 - \varepsilon$ to $0.5 + \varepsilon$. ϕ is continuous, so the integral of ϕ goes to 0 for $\varepsilon \to 0$. The δ gives -1, so the slope jumps:

$$\phi'(0.5^+) - \phi'(0.5^-) = -1.$$

Left part ($x < 0.5$): $\phi(0) = 0$ removes the cos, so $\phi = A\sin(\pi x)$.
Right part ($x > 0.5$): $\phi(1) = 0$ also removes the cos, because $\cos(\pi) = -1 \neq 0$, so $\phi = C\sin(\pi x)$.
Both have slope 0 at $x = 0.5$ because $\cos(\pi/2) = 0$. So the jump is 0, not -1. Therefore, the equation has no solution also.

4) Is the solution of the weak form the same as the solution of the equation?

For $k = \pi$ both have no solution, because of the same resonance with $\sin(\pi x)$. The weak form only needs ϕ', so it is ok with the kink from the δ (section 9.3), but the resonance is still there. So the answer is the same for both: no solution.